Arithmetic Engineering with Character-Resonators

From L-functions to Devices: Cryptography, Quantum Gates, Signal Processing, and Metamaterials

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Executive Summary

Character-Resonators form a family of physical and computational devices parameterized by characters chi. They natively recognize and react to deep arithmetic patterns tied to Dirichlet L-functions. This document states the core principle and details four high-value applications with concrete engineering roadmaps.

Core Principle

Each character chi specifies a distinct resonant structure derived from the Character-Resonator Hamiltonian H_delta(s, chi). Signal features mapped from the nontrivial zeros of L(s, chi) are used to drive devices that are selective to that character. Engineering note: for device implementations we use an ASCII-friendly mapping: omega_n = a * t_n + b, where t_n are the imaginary parts of zeros on the critical line and (a, b) set physical units (frequency or path length).

1. Provably Secure, Arithmetic-Selective Cryptography

What it is: a communication stack where the lock and key are determined by a chosen character chi. The receiver is a Character-Resonator tuned to chi; it is inert to other spectra and resonates only when driven by signals encoded with the correct arithmetic structure.

Architecture (textual block diagram):

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[Sender] -> Encoder (plaintext -> spectral lines at omega_n) -> Channel (RF/optical) ->
Chi-Selective Resonator -> Demodulator -> Plaintext Output
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Key parameters:

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- Public parameter: character chi - Private modulation: selection and weighting over omega_n
derived from L(s, chi) zeros - False-accept/false-reject targets: FAR < 1e-9, FRR < 1e-6 -
Resilience: 20 dB operation below noise via coherent integration
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30/60/90-day roadmap:

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D30: Software resonator and encoder/decoder; lab tests on synthetic channels. D60: FPGA/DSP
prototype; over-the-air demo at ISM band with narrowband pilot. D90: Hardware resonator
front-end (SAW/photonic) with chi reconfiguration; security evaluation.
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2. Quantum Gates Specialized for Number Theory

What it is: quantum gates whose evolution is governed by Hamiltonians tuned by chi, enabling native arithmetic predicates. Use cases: residue-class checks, character sums, modular phase-kickback.

Prototype concepts:

- Superconducting circuit: flux-tunable coupler implements time-dependent drive at omega_n. - Trapped-ion chain: amplitude/phase modulation scheduled by omega_n to enact character-conditioned rotations.

Milestones:

M1: Simulate chi-conditioned gates with realistic decoherence. M2: Bench-top demonstration of a single-qubit chi-check gate (fidelity > 0.99). M3: Two-qubit entangling gate incorporating chi-selective interaction.

3. Ultra-Sensitive Signal Processing for Pattern Discovery

What it is: a bank of thousands of Character-Resonators that detects faint structured signals embedded in noise. Applications: secure comms, SETI, and precision sensing.

Processing flow:

Wideband Input -> Prewhitener -> Resonator Bank {chi_k} -> Coherent Integrator -> Threshold / Decision -> ID of chi

Performance targets:

- Processing gain: > 40 dB via coherent integration across matched omega_n. - Miss probability at SNR = -15 dB: < 1e-3. - False-alarm rate per hour (1000 resonators): < 1e-2.

4. Designer Metamaterials with Arithmetic Properties

What it is: photonic/acoustic structures whose pass/stop bands are programmed by chi through geometry and material loading. Examples: ultra-selective filters, secure waveguides that conduct only chi-coded excitations.

Fabrication notes:

- Photonics: ring-resonator arrays; path-lengths proportional to omega_n mapping. - RF/microwave: coupled resonators with varactor-tuned elements encoding the omega_n grid. - Acoustics: 3D-printed Helmholtz arrays matching targeted band structure.

Summary

Character-Resonators enable devices that operate directly on arithmetic structure. They extend number theory from abstraction to engineered function and launch the discipline of Arithmetic Engineering.

Application	Key Metric	Target
Crypto	FAR / FRR	FAR < 1e-9, FRR < 1e-6
Quantum	Gate Fidelity	> 0.99 single-qubit, > 0.97 two-qubit
Signal Processing	Detect SNR	-15 dB at miss < 1e-3
Metamaterials	Out-of-band Rejection	> 80 dB

Prepared for immediate prototyping and partner discussions.